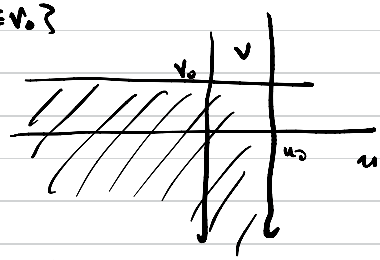
Independence of r.v.'s

X and Y are independent $\quad X \perp Y$

if $\forall X \in A,\ Y \in B$

$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\,\mathbb{P}(Y \in B)$

Let $A = \{u : u \le u_0\} \quad B = \{v : v \le v_0\}$



X and Y are independent

if and only if (iff) necessary and sufficient

$\Updownarrow$

1) $F_{X,Y}(u,v) = F_X(u)\,F_Y(v)$

2) For jointly discrete r.v.'s

$p_{X,Y}(u,v) = p_X(u)\,p_Y(v)$

3) For jointly continuous r.v.'s

$f_{X,Y}(u,v) = f_X(u)\,f_Y(v)$

$X \perp Y$ iff

$\forall u \in \mathbb{R}$ and $f_X(u) \neq 0$

$f_{Y|X}(v|u) = f_Y(v) \quad \forall v \in \mathbb{R}$

Proof

$$f_{Y|X}(v|u) = \begin{cases} \dfrac{f_{X,Y}(u,v)}{f_X(u)} & f_X(u) \neq 0 \\ \text{undefined} & f_X(u) = 0 \end{cases}$$

$$X \perp Y \iff \frac{f_{X,Y}(u,v)}{f_X(u)} = \frac{f_X(u)\,f_Y(v)}{f_X(u)} = f_Y(v)$$

Then $f_{Y|X}(v|u) = \begin{cases} f_Y(v) & \text{for } f_X(u) \neq 0 \\ \text{undefined} & \text{for } f_X(u) = 0 \end{cases}$

ex $f_{X,Y}(u,v) = \begin{cases} 6e^{-2u}e^{-3v} & \text{for } u>0,\ v>0 \\ 0 & \text{else} \end{cases}$

1) Verify this is a valid pdf

$$\int_0^\infty \int_0^\infty 6e^{-2u}e^{-3v}\,dv\,du$$

$$= \int_0^\infty \left[-2e^{-2u}e^{-3v}\Big|_0^\infty\right] du$$

$$= \int_0^\infty (0 + 2e^{-2u})\,du$$

$$= -e^{-2u}\Big|_0^\infty = 0 + 1 = 1 \quad ✓$$

2) Are X, Y independent?

$$f_{X,Y}(u,v) = f_X(u)\, f_Y(v)$$

$$f_X(u) = \int_0^\infty f_{X,Y}(u,v)\,dv = \int_0^\infty 6e^{-2u}e^{-3v}\,dv$$

$$= -2e^{-2u}e^{-3v}\Big|_0^\infty$$

$$= 0 + 2e^{-2u} = 2e^{-2u}$$

$$f_Y(v) = \int_0^\infty 6e^{-2u}e^{-3v}\,du$$

$$= -3e^{-2u}e^{-3v}\Big|_0^\infty = 0 + 3e^{-3v}$$

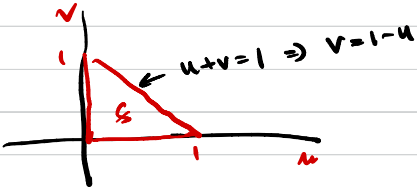
$$= 3e^{-3v}$$

$$f_X(u)\,f_Y(v) = 2e^{-2u}\cdot 3e^{-3v}$$

$$= 6e^{-2u}e^{-3v} = f_{X,Y}(u,v)$$

$$\Rightarrow X \perp Y$$

eg $f_{X,Y}(u,v) = \begin{cases} 24uv & 0<u<1,\ 0<v<1,\ 0<u+v<1 \\ 0 & \text{else} \end{cases}$



1) Verify pdf

$$\int_0^1 \int_0^{1-u} 24uv \quad dv\,du$$

$$= \int_0^1 \left[12uv^2\Big|_0^{1-u}\right] du$$

$$= \int_0^1 12u(1-u)^2\,du$$

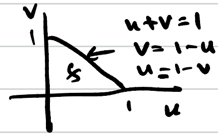
$$= \int_0^1 12u\cdot(u^2-2u+1)\,du$$

$$= \int_0^1 12u^3 - 24u^2 + 12u\,du$$

$$= 3u^4 - 8u^3 + 6u^2\Big|_0^1$$

$$= 3-8+6 = 1 \quad \checkmark$$

2) Are X, Y independent ?



$$f_X(u) = \int_S 24uv \, dv$$
$$= \int_0^{1-u} 24uv \, dv$$
$$= 12uv^2 \Big|_0^{1-u} = 12u(1-u)^2$$

$$f_Y(v) = \int_0^{1-v} 24uv \, du$$
$$= 12v(1-v)^2$$

$$f_X(u) f_Y(v) = 144 \, uv(1-u)^2(1-v)^2 \neq \underbrace{24uv}_{f_{X,Y}(u,v)}$$

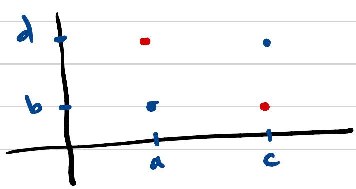
X and Y are NOT independent.

Swap Property

A subset $S \subset \mathbb{R}^2$ has the swap property if

$\forall (a,b) \in S$ and $\forall (c,d) \in S$

(a,d) and (c,b) also $\in S$



Product Sets

Let $S \subset \mathbb{R}^2$

Then S is a product set

iff $\Updownarrow$

S has the swap property

If X, Y are independent jointly continuous r.v.'s
$\Downarrow$ (necessary condition)
Then support S of $f_{X,Y}(u,v)$ is a swap set

If S is not a swap set
$\Downarrow$
Then X, Y not independent

* If S is a swap set
X, Y are not necessarily independent

Distribution of sum of r.v.'s

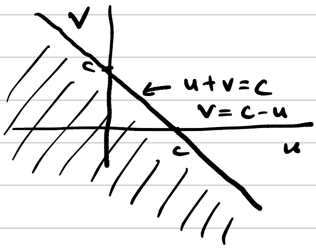
Let X, Y be jointly continuous r.v.'s $(f_{X,Y}(u,v))$

And let $Z = X + Y$
Find $f_Z(c)$

$$\underbrace{F_Z(c)}_{\text{CDF}} = \mathbb{P}(Z \le c) = \mathbb{P}(X + Y \le c)$$

$$= \mathbb{P}(Y \le c - X)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{c-u} f_{X,Y}(u,v)\, dv\, du$$



pdf $f_Z(c) = \frac{d}{dc} F_Z(c)$

$$= \frac{d}{dc} \int_{-\infty}^{\infty} \int_{-\infty}^{c-u} f_{X,Y}(u,v)\, dv\, du$$

$$= \int_{-\infty}^{\infty} \frac{d}{dc} \left[\int_{-\infty}^{c-u} f_{X,Y}(u,v)\, dv \right] du$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(u, c-u)\, du \leftarrow$$

If $X \perp Y$ $\quad f_{X,Y}(u,v) = f_X(u) f_Y(v)$

$$f_{X,Y}(u, c-u) = f_X(u) f_Y(c-u)$$

$$f_Z(c) = \int_{-\infty}^{\infty} f_X(u) f_Y(c-u)\, du \quad \leftarrow \quad u \to \tau, \; c \to t$$

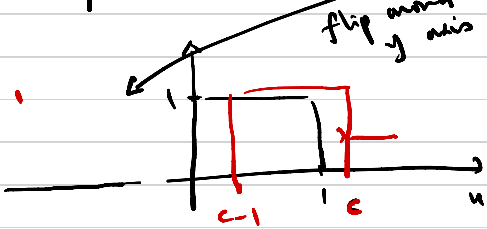
$$\int_{-\infty}^{\infty} f_X(\tau) f_Y(t-\tau)\, d\tau \quad \leftarrow \text{convolution}$$

$$f_Z = f_X * f_Y \qquad Z = X+Y \qquad X \perp Y$$

ex Let $X, Y \sim \text{Uniform}(0,1)$ i.i.d. independent identically distributed

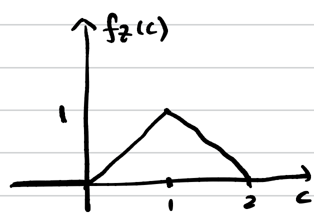
Let $Z = X + Y$

Find $f_Z(c)$



$1-(c-1) = 2-c$

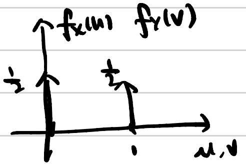
$$f_Z(c) = \begin{cases} 0 & c \le 0 \\ c & c \in [0,1] \\ 2-c & c \in [1,2] \\ 0 & c > 2 \end{cases}$$

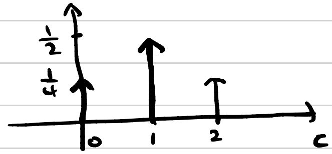


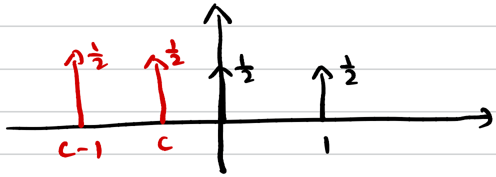
eg

Let $X, Y \sim \text{Bernoulli}(\tfrac{1}{2})$ i.i.d.

$Z = X + Y$ $\qquad Z \sim \text{Binomial}(2, \tfrac{1}{2})$







Discrete convolution for Pmf

$$P_Z(k) = \sum_j P_X(j) P_Y(k-j)$$

eg

$X \sim \text{Poisson}(\lambda_1)$ $\qquad X \perp Y$

$Y \sim \text{Poisson}(\lambda_2)$

$Z = X + Y$ $\qquad$ Find $P_Z(k)$

$$P_X(k) = \frac{\lambda_1^k e^{-\lambda_1}}{k!} \qquad k \geq 0$$

$$P_Y(k) = \frac{\lambda_2^k e^{-\lambda_2}}{k!} \qquad k \geq 0$$

$$P_Z(k) = \sum_{j=0}^{k} \frac{\lambda_1^j e^{-\lambda_1}}{j!} \frac{\lambda_2^{k-j} e^{-\lambda_2}}{(k-j)!}$$

$$= e^{-(\lambda_1+\lambda_2)} \sum_{j=0}^{k} \frac{\lambda_1^j \lambda_2^{k-j}}{j!(k-j)!}$$

$$\frac{e^{-\lambda}}{k!} \sum_{j=0}^{k} \frac{\lambda^j p^j \lambda^{(k-j)} (1-p)^{(k-j)}}{j!(k-j)!}$$

Binomial pmf.

$\binom{n}{i} p^i (1-p)^{n-i}$

$\frac{n!}{i(n-i)!} p^i (1-p)^{n-i}$

Let $\lambda = \lambda_1 + \lambda_2$

$p = \frac{\lambda_1}{\lambda} \quad 1-p = \frac{\lambda_2}{\lambda}$

$n = k \quad i = j$

$$= \frac{\lambda^k e^{-\lambda}}{k!} \underbrace{\sum_{j=0}^{k} \binom{k}{j} p^j (1-p)^{k-j}}_{1}$$

$$= \frac{\lambda^k e^{-\lambda}}{k!} \qquad \lambda = \lambda_1 + \lambda_2$$

$Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$

eg $\quad X \sim N(\mu_1, \sigma_1^2) \qquad X \perp Y$

$Y \sim N(\mu_2, \sigma_2^2)$

$Z = X + Y \qquad Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Find $f_Z(c) = f_X(u) * f_Y(v)$

$$f_X(u) = \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{\left[-\frac{(u-\mu_1)^2}{2\sigma_1^2}\right]}$$

$$*$$

$$f_Y(v) = \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{\left[-\frac{(v-\mu_2)^2}{2\sigma_2^2}\right]}$$

$$f_X(u) * f_Y(v) = \int_{-\infty}^{\infty} f_X(u) f_Y(c-u)\, du$$

$$f_X(t) * f_Y(t) \xleftrightarrow{F.T.} \mathcal{F}(f_X)\mathcal{F}(f_Y)$$

$$e^{\frac{-t^2}{2\sigma^2}} \longleftrightarrow \sigma\sqrt{2\pi}\, e^{-\frac{\sigma^2 w^2}{2}}$$

Let $X_0 = X - \mu_1 \qquad X_0 \sim N(0, \sigma_1^2)$

$Y_0 = Y - \mu_2 \qquad Y_0 \sim N(0, \sigma_2^2)$

$$Z = \underbrace{X_0 + Y_0}_{Z_0} + \mu_1 + \mu_2$$

$$f_{X_0}(u) = \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{\frac{-u^2}{2\sigma_1^2}} * f_{Y_0}(v) = \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{\frac{-v^2}{2\sigma_2^2}}$$

$\mathcal{F}$

$$\mathcal{F}(f_{X_0}) = \underbrace{\cancel{\frac{1}{\sqrt{2\pi\sigma_1^2}}\sigma_1\sqrt{2\pi}}}_{=1} e^{-\frac{\sigma_1^2 w^2}{2}} = e^{-\frac{\sigma_1^2 w^2}{2}}$$

$$\frac{1}{\sigma_1\sqrt{2\pi}}$$

$$\mathcal{F}(f_{Y_0}) = \underbrace{\cancel{\frac{1}{\sigma_2\sqrt{2\pi}}\sigma_2\sqrt{2\pi}}}_{1} e^{-\frac{\sigma_2^2 w^2}{2}}$$

$$f_{X_0}(u) * f_{Y_0}(v) = \mathcal{F}^{-1}\{\mathcal{F}(f_{X_0})\mathcal{F}(f_{Y_0})\}$$

$$= \mathcal{F}^{-1}\{e^{-\frac{\sigma_1^2 w^2}{2}} e^{-\frac{\sigma_2^2 w^2}{2}}\} = \mathcal{F}^{-1}\{e^{-\frac{(\sigma_1^2+\sigma_2^2)w^2}{2}}\}$$

$$\downarrow$$

$$e^{\frac{-t^2}{2\sigma^2}} \leftrightarrow \sigma\sqrt{2\pi}\, e^{\frac{-\sigma^2\omega^2}{2}} \qquad = \frac{1}{\sqrt{2\pi(\sigma_1^2+\sigma_2^2)}}\, e^{\frac{-t^2}{2(\sigma_1^2+\sigma_2^2)}} = f_{Z_0}(t)$$

$$Z_0 = X + Y \qquad Z_0 \sim \mathcal{N}(0, \sigma_1^2+\sigma_2^2)$$

$$Z = Z_0 + \mu_1 + \mu_2$$

$$Z \sim \mathcal{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

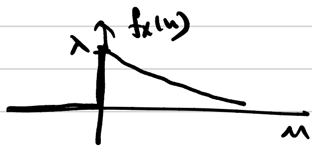
<u>Find pdf of function of two r.v.'s</u>

<u>ex</u> $X, Y \sim \mathrm{Exp}(\lambda)$ i.i.d.

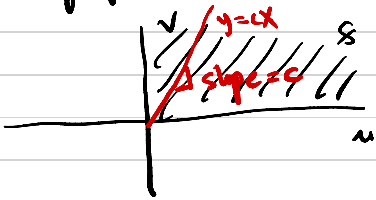
$R = \frac{Y}{X}$ find f_R

$f_X(u) = \lambda e^{-\lambda u}$ $u \in [0,\infty)$

$f_Y(v) = \lambda e^{-\lambda v}$ $v \in [0,\infty)$



Support of $f_{X,Y}$: $[0,\infty) \times [0,\infty)$



$f_{X,Y}(u,v) = f_X(u)\, f_Y(v)$ ← $X \perp Y$

$= \lambda^2 e^{-\lambda(u+v)}$ $(x,y) \in [0,\infty) \times [0,\infty)$

Support of R : $[0,\infty)$

$$F_R(c) = \mathbb{P}(R \le c) = \mathbb{P}\left(\frac{Y}{X} \le c\right)$$

$$= \mathbb{P}(Y \le cX)$$

$$= \int_0^\infty \int_0^{cu} \lambda^2 e^{-\lambda(u+v)}\, dv\, du \qquad \lambda^2 e^{-\lambda u} e^{-\lambda v}$$

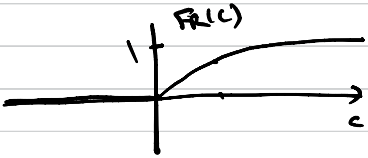
$$= \int_0^\infty -\lambda e^{-\lambda u} e^{-\lambda v} \Big|_0^{cu}\, du$$

$$= \int_0^\infty \underbrace{-\lambda e^{-\lambda u} e^{-\lambda c u}}_{-\lambda e^{-\lambda(1+c)u}} + \lambda e^{-\lambda u}\, du$$

$$= \frac{1}{1+c} e^{-\lambda(1+c)u} - e^{-\lambda u} \Big|_0^\infty$$

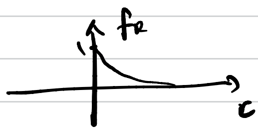
$$= 0 - 0 - \frac{1}{1+c} + 1$$

$$F_R(c) = 1 - \frac{1}{1+c} = \frac{1+c-1}{1+c} = \begin{cases} \frac{c}{1+c} & c \geq 0 \\ 0 & \text{else} \end{cases}$$



$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$f_R(c) = \frac{d}{dc} F_R(c) = \frac{1+c-c}{(1+c)^2} = \begin{cases} \frac{1}{(1+c)^2} & c \geq 0 \\ 0 & \text{else} \end{cases}$$



<u>eg</u> $X, Y \sim Exp(\lambda)$ i.i.d. $f_X(u) = \lambda e^{-\lambda u}$

$G = \frac{X}{X+Y}$ find f_G

$$f_{X,Y}(u,v) = \begin{cases} \lambda^2 e^{-\lambda u} e^{-\lambda v} & (u,v) \in [0,\infty) \times [0,\infty) \\ 0 & \text{else} \end{cases}$$

Support of G: $[0,1]$

$$F_G(c) = \mathbb{P}(G \leq c) = \mathbb{P}\left(\frac{X}{X+Y} \leq c\right)$$

$$= \mathbb{P}(X \leq cX + cY)$$

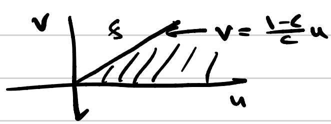
$$= \mathbb{P}((1-c)X \leq cY)$$

$$= \begin{cases} \mathbb{P}\left(\frac{1-c}{c} X \leq Y\right) & c \geq 0 \\ \underbrace{\mathbb{P}\left(\frac{1-c}{c} X \geq Y\right)}_{=0} & c < 0 \end{cases}$$

cannot happen outside support of G

$$F_G(c) = \mathbb{P}\left(Y \geq \frac{1-c}{c} X\right)$$

$$= 1 - \mathbb{P}\left(Y < \frac{1-c}{c} X\right)$$



$$= 1 - \int_0^\infty \int_0^{\frac{1-c}{c} u} f_{X,Y}(u,v) \, dv \, du$$

$$= 1 - \frac{c'}{1+c'}$$

$$= 1 - \frac{1-c}{\cancel{c}}\,\cancel{c}$$

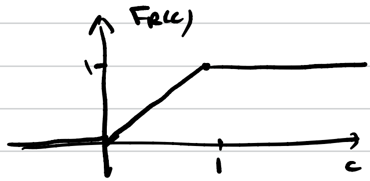
$$= 1 - 1 + c = c$$

$$c' = \frac{1-c}{c}$$

$$1+c' = \frac{c+1-c}{c} = \frac{1}{c}$$

$$= \int_0^{\infty}\int_0^{cu} \lambda^2 e^{-\lambda(u+v)}\, dv\, du = \frac{c}{1+c}$$

$$F_R(c) = \begin{cases} c & c \in [0,1] \\ 0 & c < 0 \\ 1 & c > 1 \end{cases}$$

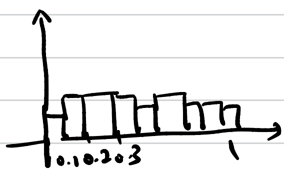


$$f_R(c) = \frac{d}{dc} F_R = \begin{cases} 1 & \in [0,1] \\ 0 & \text{else} \end{cases}$$

$$G = \frac{X}{X+Y} \qquad G \sim \text{Uniform}(0,1)$$

$X, Y \sim \text{Exp}(\lambda)$ i.i.d.

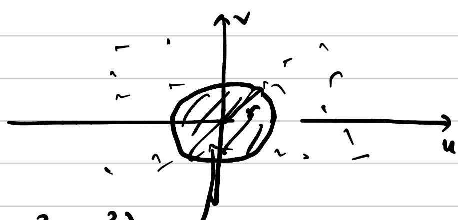
```
X = rand(10000, 'Exp', λ)
Y = rand(10000, 'Exp', λ)
G = X/(X+Y)
hist(G)
```



ex Given $f_{X,Y}(u,v)$

$$R = \sqrt{X^2 + Y^2}$$

Find $f_R(r)$



$$F_R(r) = \mathbb{P}(R \le r) = \mathbb{P}(X^2 + Y^2 \le r^2)$$

$$= \int_0^{2\pi}\int_0^{r} f_{X,Y}(r'\cos\theta, r'\sin\theta)\, r'\, dr'\, d\theta$$

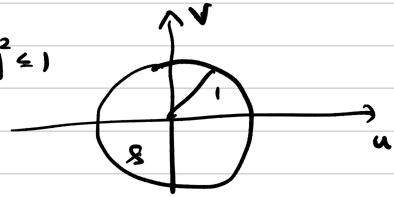
$u = r'\cos\theta$
$v = r'\sin\theta$

$$f_R(r) = \frac{d}{dr} F_R(r)$$

$$= \int_0^{2\pi} \frac{d}{dr}\left[\int_0^r f_{X,Y}(r'\cos\theta, r'\sin\theta)\, r'\,dr'\right] d\theta$$

$$= \int_0^{2\pi} f_{X,Y}(r\cos\theta, r\sin\theta)\, r\, d\theta \quad \leftarrow$$

eg If $f_{X,Y}(u,v) = \begin{cases} \frac{1}{\pi} & 0 \le x^2+y^2 \le 1 \\ 0 & \text{else} \end{cases}$



$\pi r^2 = \pi$

$$f_R(r) = \int_0^{2\pi} \frac{1}{\pi} r\, d\theta \qquad 0 \le r^2 \le 1$$

$$= \begin{cases} 2r & 0 \le r \le 1 \\ 0 & \text{else} \end{cases}$$

eg Given $f_{X,Y}(u,v)$

Find $f_{R,\Theta}(r,\theta)$ where $R = \sqrt{X^2+Y^2}$, $\Theta = \tan^{-1}\frac{Y}{X}$

R, Θ prob. not indep.

$g(u,v) = (r,\theta)$ $\qquad g(u,v) = \left(\sqrt{u^2+v^2}, \tan^{-1}\frac{v}{u}\right)$

$(u,v) = g^{-1}(r,\theta)$ $\qquad$ polar → cartesian

$$\begin{pmatrix} u = r\cos\theta \\ v = r\sin\theta \end{pmatrix} = g^{-1}(:) \quad \leftarrow$$

Find $f_{W,Z}$ from $f_{X,Y}$ when $\binom{W}{Z} = g\binom{X}{Y}$

For $\binom{W}{Z} = g\left(\binom{X}{Y}\right)$ $\qquad f_{X,Y}$ is known

$$f_{W,Z}(\alpha,\beta) = \frac{1}{|\det \bar{J}|} f_{X,Y}\left(g^{-1}\left(\begin{pmatrix}\alpha\\ \beta\end{pmatrix}\right)\right) \qquad g = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}$$

$$\bar{J} = \text{Jacobian} = \begin{bmatrix} \frac{\partial}{\partial u} g_1(u,v) & \frac{\partial}{\partial v} g_1(u,v) \\ \frac{\partial}{\partial u} g_2(u,v) & \frac{\partial}{\partial v} g_2(u,v) \end{bmatrix}$$

$g_1(X,Y) = W$

$g_2(X,Y) = Z$